Alfred Biju

LinkedIn: www.linkedin.com/in/alfredbiju13
GitHub: https://github.com/alfredd25
Email: alfredbiju7@gmail.com
Mobile: +91 90747 58241

Skills

- **Languages**: Python, SQL
- **Backend & APIs**: FastAPI, Flask, REST APIs, WebSockets, AsyncIO, SQLAlchemy
- **Tools/Devops:** Git, Docker, Jenkins, AWS (EC2, EBS, S3), Nginx, Celery, Prometheus, Grafana, Alembic, Pytest
- **Databases**: PostgreSQL, Redis, MySQL, MongoDB
- **Soft Skills**: Ownership, Structured Problem Solving, Technical Communication, Cross-team collaboration

Experience

Freelancer.com – Python developer (Project-Based) | Profile Since Dec '25
Remote

- Engineered scalable backend APIs using Python and FastAPI, handling 1,000+ API requests per day with optimized routing and structured error handling
- Architected and developed Telegram bot systems integrating REST APIs and databases, serving 38,000+ active user group
- Executed asynchronous processing using async/await, reducing response time by ~30% and improving system performance under concurrent requests

Projects

- **NutriTrack – Full-Stack Calorie Tracker | GitHub** Mar '26 – Present
  - Built a production-style FastAPI backend with PostgreSQL, SQLAlchemy, and Alembic, enabling secure APIs and 100% migration-driven schema management
  - Implemented Celery + Redis workflows for USDA imports, daily summaries, and scheduled jobs, improving API responsiveness by ~30% under load
  - Orchestrated 6+ services with Docker Compose, integrating FastAPI, PostgreSQL, Redis, Nginx, Prometheus, and Grafana, and reducing setup effort by ~70%.
  - Automated Jenkins CI/CD pipeline to run containerized pytest suites, publish Docker images, and deploy to AWS EC2, cutting manual deployment effort by ~80%
  - **Tech Stack:** Python, FastAPI, Next.js, React, PostgreSQL, Redis, Celery, Docker, AWS EC2, Jenkins, Prometheus, Grafana, SQLAlchemy, Nginx

- **Crypto Matching Engine | GitHub** Sep '25 – Oct '25
  - Built a high-performance cryptocurrency order matching engine implementing strict price-time priority matching with O(log n) price level lookups using SortedDict
  - Designed dual-sided order book with deque-based FIFO queues and Decimal precision arithmetic to ensure sub-penny accuracy across millions of orders
  - Implemented 4 advanced order types (Market, Limit, IOC, FOK) with internal trade-through protection and partial fill support; validated 5,000+ order throughput with load testing
  - Exposed REST and WebSocket APIs using FastAPI/Uvicorn for real-time order book snapshots and trade streaming with 35+ comprehensive pytest unit and integration tests (100% critical path coverage)
  - **Tech Stack:** Python, FastAPI, uvicorn, pydantic, pytest, websockets

Achievements

- **Hackathon Finalist – Fynd Hactimus** : Dec '25

  Built a production-ready solution using Python, FastAPI, and Boltic; selected among top finalists out of 140+ teams.
- **Data Structures & Algorithms Practice**: Jan '24 - Present

  Solved 200+ coding problems across various coding platforms focusing on Trees, Graphs, Dynamic Programming, and core problem-solving patterns.

Education

Lovely Professional University Punjab, India
Bachelor of Technology - Computer Science and Engineering; ***CGPA: 7.8*** *Since August 2023*

Lisieux English School Vaikom, Kerala
Intermediate; ***Percentage 85.6%*** *April 2021 - March 2023*

Lisieux English School Vaikom, Kerala
Matriculation; ***Percentage 93%*** *April 2020 - March 2021*